Multi-Agent Retrieval as an Alternative to Classical RAG: A Controlled Comparison on Educational Tutoring Benchmarks

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Abstract— Classical retrieval-augmented generation (RAG) treats retrieval as the top-level controller: a user query is embedded, k passages are fetched, and a reader generates from them. In educational tutoring, this abstraction can be limiting, because the right passages depend on the learner's state–prior level, misconception, rubric–not just the surface query. We test whether a multi-agent coordination stack can replace classical RAG as the grounding mechanism. The comparison has four retrieval mechanisms with the same open-weight backbone, corpus, reranker, tutor, critic, and response budget: (i) classical RAG, (ii) retrieval-free multi-agent, (iii) a single-agent no-retrieval floor, and (iv) our multi-agent retrieval mechanism (hybrid_fast), which runs planner, diagnoser, and rubric agents before deciding whether retrieval is needed. On TutorEval [1] (n=834 identical examples), multi-agent retrieval beats classical RAG on Token-F1 (+0.004, p=0.04), tutor-hit (+0.019, p=0.005), rubric coverage (+0.026, p=0.006), and total tokens (−484, $p<10^{-4}$). On the EduBench [2] intersection (n=1562 identical example_ids), the gap is larger: EB12D +0.031, rubric +0.186, latency −5.9 s, tokens −45%, and LLM calls −25%, all at $p<10^{-4}$. The LLM-call result directly addresses the capacity-confound objection: on EduBench, our coordination-heavy pipeline uses fewer backbone calls than retrieve-then-read. On Qwen3.6-27B-FP8, TutorEval quality saturates for both systems, but the efficiency gap widens; on EduBench, the quality and efficiency gains both persist. Multi-agent coordination is therefore a practical replacement for the retrieve-first controller in classical RAG: it preserves retrieval when external evidence is useful, but subordinates retrieval to a routed context policy.

I. Introduction

Classical retrieval-augmented generation (RAG) has become the default mechanism for grounding language-model outputs in external evidence [3]. Its pipeline is fixed by design: the user query is embedded, top-k passages are fetched from an index, and a reader conditions on those passages to generate an answer. This pipeline is optimized for query-determined evidence–the retrieved chunks are a function of the surface text of the query. Educational tutoring breaks that assumption. What a tutor should say to a student who asks "please explain phase to me" depends not only on which textbook chunk mentions "phase" but on the learner's prior level, the rubric the grader will use, and the misconception most likely behind the question. When the retrieval target depends on learner state, embedding the raw query directly is the wrong abstraction.

A natural alternative is to let a small group of specialist agents do the work that classical RAG assumes a single embedded query can do. A planner can propose a scoped search intent; a diagnoser can infer the learner's level and likely misconception; a rubric agent can compile the criteria the answer will be graded against; and only then does a retriever (if anyone) consult the corpus. We refer to this family of systems as multi-agent retrieval. Unlike classical RAG, the retriever is no longer the top-level controller; it becomes an optional tool inside a coordination policy that has already built a representation of what the learner needs.

This paper asks whether multi-agent retrieval can replace classical RAG as the grounding mechanism in educational tutoring. We formulate the comparison as a controlled ablation over four retrieval mechanisms that share the same Qwen3.5-4B backbone (thinking budget 512), the same shared corpus, the same reranker, and the same response budget: (i) Classical RAG (retrieve-always, retrieve-then-read), (ii) Multi-Agent (no retrieval), which replaces the retriever with parallel planner, diagnoser, and rubric agents, (iii) Single-Agent (no retrieval) as an ablation floor, and (iv) our Multi-Agent Retrieval mechanism (hybrid_fast), in which the coordination agents run first and a lightweight router then decides per example whether the retriever should be called. The contrast is deliberate: all four mechanisms see the same prompt; the only thing that varies is who–or what–decides when and whether the corpus is consulted.

Two education benchmarks give us the regime variety we need: they make the source of useful grounding context ambiguous, ranging from external instructional evidence to learner-, rubric-, and scenario-conditioned context. TutorEval [1], [4] stresses the evidence-grounded regime, where the right passage usually exists inside a textbook chunk. EduBench [2] spans grading, emotional support, and personalized-learning scenarios where the useful context depends more on learner profile and rubric than on a single chunk. Related educational-evaluation work [5]–[10] reinforces the methodological point: a grounding mechanism cannot be judged on retrieval recall alone. On TutorEval (n=834 identical examples) and on the EduBench intersection (n=1562 identical example_ids), the same tutor scores higher when fed multi-agent retrieval context while using fewer tokens; on EduBench it also reduces latency and LLM calls. This makes the education setting a useful stress test for retrieval control, because not every example is best grounded by the same kind of context.

Contributions. (i) We formalize four retrieval mech-

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anisms that differ only in how evidence is supplied to a shared reader under a common budget. (ii) We operationalize them in an open framework where the retrieval context is the only moving part: planner, diagnoser, rubric, router, and retriever vary; the tutor and critic are held constant. Released configs pin Qwen3.5-4B and Qwen3.6-27B-FP8; scripts reproduce every cell. (iii) On TutorEval (n=834) and EduBench (n=1562 intersection), paired bootstrap CIs and permutation tests show that multi-agent retrieval beats classical RAG on fixed-tutor outcome and efficiency metrics; on EduBench it also uses fewer LLM calls per example.

II. Related Work

Our work sits at the intersection of three lines: classical RAG and its extensions, agentic/multi-agent retrieval, and educational tutoring benchmarks that are sensitive to the grounding mechanism. We survey each line with an eye on one question–what serves the role classical RAG defines as "the retriever"?–because that is the question our experiments answer.

Classical RAG and retrieval-control extensions. Classical RAG fixes a retrieve-then-read controller: embed the query, retrieve top-k passages, and condition the reader on the retrieved context [3]. A growing body of work relaxes this fixed controller. Self-RAG trains reflection tokens to decide when to retrieve and critique generations [11]; CRAG inserts a retrieval evaluator that triggers corrective actions on low-confidence passages [12]; Adaptive-RAG learns a classifier over query complexity to choose among no-retrieval, single-step retrieval, and multi-step retrieval strategies [13]; and RetrievalQA audits when retrieval helps short-form QA [14]. Evaluation and long-context work further show that strong retrieval alone does not guarantee grounded generation [15]–[20]. These variants show that fixed retrieve-then-read RAG is an active target of improvement. Our distinction is that the retrieval decision is driven by an education-specific regime signal produced before retrieval from planner, diagnosis, and rubric context, rather than only by the query, retrieved passages, or a task-general learned controller. This makes our policy training-free and context-specific. We treat these related methods as complementary rather than direct experimental baselines: instantiating them would introduce additional control components or training objectives, whereas our experiment isolates the retrieve-first controller under a shared backbone, corpus, reranker, reader/critic, and budget.

Agentic and multi-agent retrieval. A parallel line treats retrieval as one tool inside a larger agentic loop. ReAct interleaves reasoning and acting so the model can decide when to call a search tool [21], [22]; AutoGen generalizes this into multi-agent collaboration with role specialization [22]; CAMEL and MetaGPT push role decomposition further [23], [24]. MA-RAG and related coordination studies show that decomposing RAG across multiple agents improves performance on tasks that require planning and staged reasoning [25], [26], and AgentBench documents how agentic control affects downstream task performance more broadly [27]. These systems implicitly argue that a coordination stack can perform the role classical RAG assigns to the retriever; we make the argument explicit via a head-to-head comparison under a shared budget.

TABLE I

Positioning of multi-agent retrieval (hybrid_fast) relative to classical RAG and its control-policy extensions. TF: training-free gate. TSC: tutoring-specific coordination of the retrieval context (rubric + diagnoser + planner assembled before the retriever, not the downstream tutor or critic). PEA: per-example activation of retrieval. COR: coordination agents run before retrieval to scope the evidence need. ✓= yes, ∼ = partial, × = no.

Retrieval mechanism	TF	TSC	PEA	COR
Classical RAG [3]	✓	×	×	×
Self-RAG [11]	×	×	✓	×
CRAG [12]	∼	×	✓	×
Adaptive-RAG [13]	×	×	✓	×
MA-RAG [25]	✓	×	✓	✓
Multi-Agent Retrieval (ours)	✓	✓	✓	✓

Educational tutoring benchmarks. Education-focused LLM evaluation has moved beyond short-form answer accuracy toward pedagogy, adaptivity, and instructional safety. EduBench spans nine tutoring scenarios and scores educational utility dimensions such as adaptation and pedagogical application [2]. TutorEval targets chapter-grounded science tutoring with keypoint-based evaluation [1], [4]. MathTutorBench, TutorBench, and related evaluation frameworks sharpen open-ended pedagogical assessment and tutor-specific safety criteria [5]–[10]. These benchmarks make the methodological point we lean on: when evaluation measures rubric adherence and calibrated help, a system's grounding mechanism cannot be judged on retrieval recall alone–because a correct chunk can still lose rubric coverage if the reader does not know the learner. Table I summarizes how our multi-agent retrieval gate relates to the RAG-extension line: it is training-free like classical RAG, activates per example like Self-RAG and CRAG, but–unlike those–the retrieval context itself is assembled by a tutoring-specific coordination stack (rubric, diagnosis, planner) that runs before the retrieval decision, leaving the downstream tutor and critic unchanged.

Multi-agent tutoring and positioning. Recent tutoring systems instantiate specialized educational roles–AgentTutor [28], IntelliCode [29], CogEvo-Edu [30]–and present themselves as agentic tutors, but from a grounding standpoint they are also candidate replacements for classical RAG: the diagnoser and planner produce the context the reader eventually generates from, even when no explicit retriever is involved. What is missing–and what we contribute–is a head-to-head comparison that treats retrieval, coordination, and their composition as alternative grounding mechanisms under

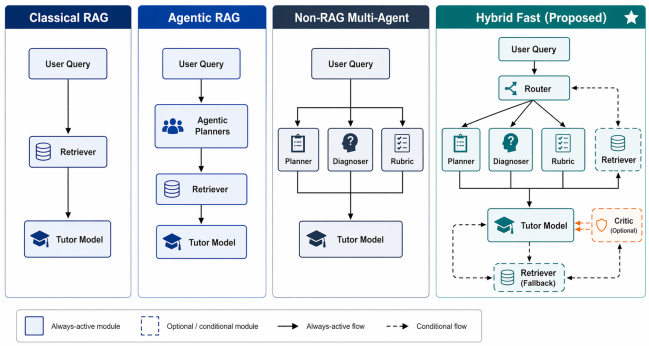

Fig. 1. The four retrieval mechanisms compared in this paper. Classical RAG places the retriever at the top level (retrieve-then-read). Multi-Agent (no retrieval) replaces the retriever with a planner/diagnoser/rubric coordination stack. Single-Agent is the ablation floor. Our Multi-Agent Retrieval (hybrid_fast) runs the coordination agents first and then gates retrieval per example, with a single fallback pass when the first attempt returns no chunks.

shared infrastructure, a frozen backbone, and statistical testing.

III. Four Retrieval Mechanisms Under a Common Budget

All four mechanisms share the same backbone, corpus, reranker, MMR packer, and response cap; the only dimension that varies is how evidence is supplied to the reader. A lightweight router selects the mechanism and sets per-example execution flags (Fig. 1). An implemented agentic_rag path is excluded because parity-controlled outputs were unavailable. The design principle is controlled adaptivity: invoke retrieval only when an instance needs external evidence, and spend the saved budget on retrieval-context coordination.

A. Regime-aware routing

For each example x, the router computes five heuristic scores $s(x) \in [0,1]^5$ over normalized keyword cues (evidence, coordination, rubric, planning, tutoring), with small dataset and context-signal bonuses (documented in regime_router.py: +0.08 for non-trivial context passages, +0.05 per attached image, +0.20 when a rubric is present, +0.12–0.18 per-dataset priors). An optional logistic-regression classifier can override the heuristic family when a trained router is loaded; our frozen runs use the heuristic path throughout. The selected mechanism is

$$\mathrm{mech}(x) = \begin{cases} \text{CLASSICAL} & s_e(x) \geq \tau_e^{\mathrm{cls}},\ s_c(x) < \tau_c \\ \text{NON-RAG} & s_c(x) \geq \tau_c,\ s_e(x) < 0.28 \\ \text{HYB-FAST} & \text{otherwise,} \end{cases} \quad (1)$$

with $(\tau_e^{\mathrm{cls}}, \tau_c) = (0.52, 0.50)$. Within hybrid_fast, the retrieval gate is

$$\mathrm{require_retrieval}(x) = \mathbb{1}\left[s_e(x) \geq \tau_e^{\mathrm{gate}} \ \vee\ \mathrm{regime}(x) = \mathrm{EG}\right], \quad (2)$$

with $\tau_e^{\mathrm{gate}} = 0.35$ and EG the evidence-grounded regime. A fallback retrieval pass fires when the draft has no chunks but $s_e(x) \geq \tau_e^{\mathrm{fall}} = 0.45$. All thresholds are the production values shipped with the repository (RouterConfig); a sweep $\tau_e^{\mathrm{gate}} \in [0.30, 0.55]$ produced by run_threshold_sweep.py shows quality is flat on $[0.30, 0.45]$.

B. Specialist agents and shared context

All four retrieval mechanisms operate on a shared AgentContext carrying the example, route decision, student state, search queries, rubric summary, retrieved chunks, and draft answer. The tutor and the critic are held constant across all four mechanisms: the tutor drafts from whatever context is supplied; an optional critic revises only on concrete issues (missing evidence markers, unmet rubric items, length mismatch). The only things we vary are the agents that build the retrieval context before the tutor is invoked: the planner emits up to three search queries tailored to the regime; the diagnoser infers learner level, goals, style, and likely misconceptions; the rubric agent compiles a checklist from rubric items; the router gates the retriever per example. Every agent shares the same small-model backbone and budget policy, so differences reflect retrieval-context choices, not tutor or capacity changes.

C. Hybrid retrieval, reranking, and context packing

The retrieval stack is intentionally lightweight. A hybrid index combines word-level TF–IDF ($\lambda_w = 0.58$), character n-gram TF–IDF ($\lambda_c = 0.17$), and an optional truncated-SVD latent projection ($\lambda_\ell = 0.25$); the weighted similarity is boosted by query–title token overlap. A six-feature logistic reranker (base score, token/title overlap, phrase match, numeric overlap, brevity) reorders the candidates, and a maximum marginal relevance [31] packer with $\lambda_{\mathrm{MMR}} = 0.68$ selects up to four chunks within a 6,000-character context budget.

D. Evaluation protocol

Metric handling is profile-aware and enforced in code: shared metrics include composite score, Token-F1, rubric coverage, latency, token count, and LLM-call count; EduBench additionally reports its 12-dimension educational rubric and score alignment, TutorEval reports tutor-hit and chapter-grounded tutoring quality. Profile-inapplicable metrics are omitted from the headline comparison rather than forced to zero. A retrieval-agnostic grounding proxy scores answer sentences against the shared corpus index regardless of whether the pipeline invoked retrieval.

IV. Experimental Results

We run two frozen head-to-head comparisons under matched infrastructure (Qwen3.5-4B, thinking budget 512, temperature 0.15, response cap 900 tokens). TutorEval is complete on n=834 identical examples across the four mechanisms; EduBench is evaluated on the intersection of n=1562 identical example_ids processed by all four mechanisms, so every claim compares point

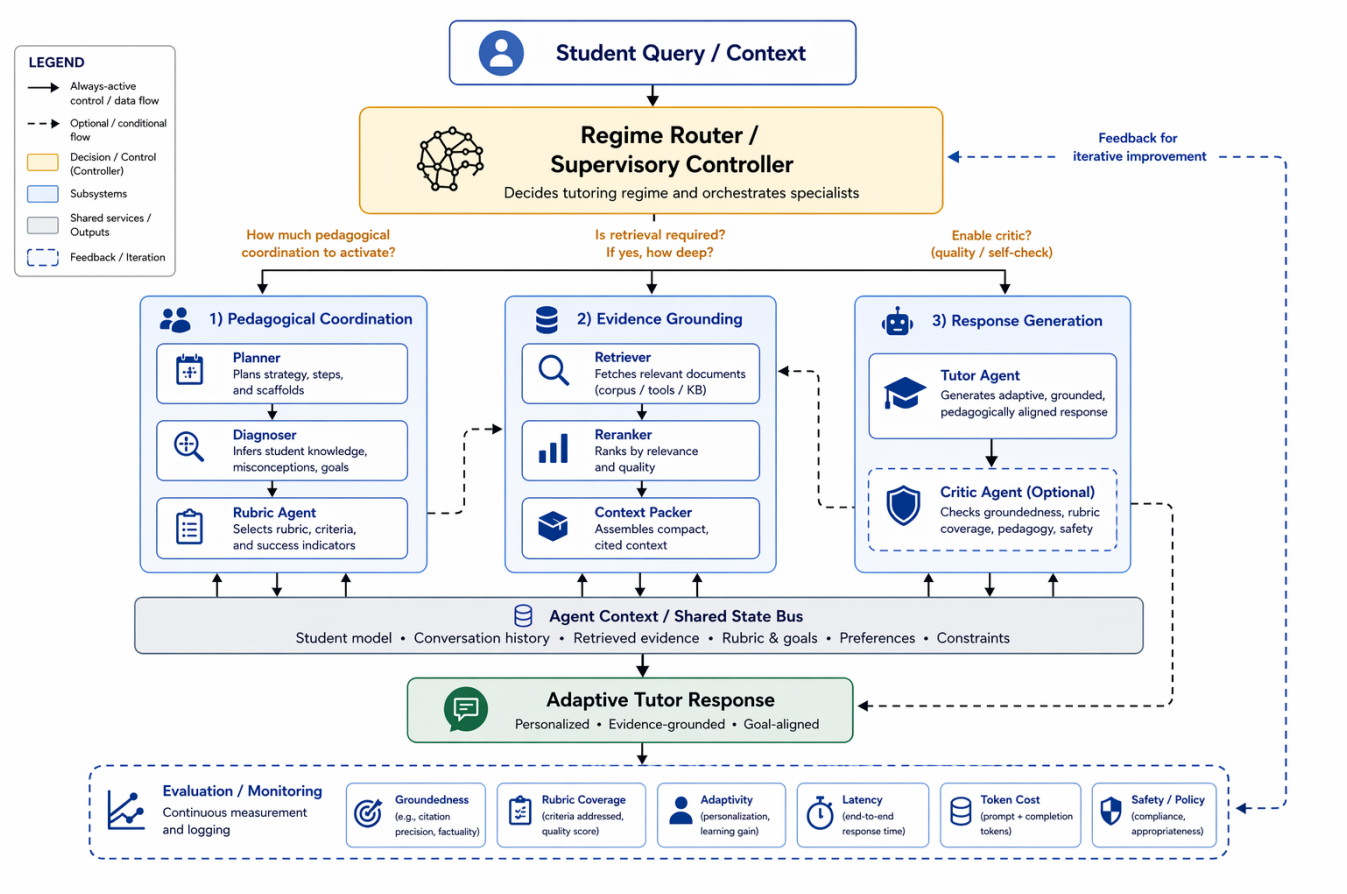

Fig. 2. Full pipeline view of the proposed Multi-Agent Retrieval mechanism (hybrid_fast). Planner, diagnoser, and rubric agents run in parallel to build a shared context; a lightweight router then gates the retriever per example. Retrieval, when triggered, uses the planner's scoped queries; a single fallback retrieval pass is issued when the tutor's draft has no grounding but the evidence score remains high. Unlike classical RAG, the retriever is a conditional tool inside the coordination policy, not the top-level controller.

estimates computed on the same test set. Retrieval hyperparameters and budget caps are fixed at the §III values.

A. Primary comparison: TutorEval

Table II reports the main comparison with paired bootstrap 95% CIs and permutation p-values for each (hybrid, classical) delta. On TutorEval, changing the retrieval context improves the shared tutor's reported outcomes while using fewer tokens: Token-F1 +0.004 (p=0.04), tutor-hit +0.019 (p=0.005), rubric coverage +0.026 (p=0.006), and tokens −484 ($p<10^{-4}$, −12%). Latency is directionally lower but not significant (−2,790 ms, p=0.21). The retrieval gate fires on every TutorEval example (100% of 834): chapter-grounded science questions keep the evidence score s_e high, so hybrid is classical RAG plus coordination here, not a strict subset of it. LLM calls therefore rise to 3.96 vs. 2.01, but the tutor and critic remain shared with classical RAG. Retrieval-free multi-agent and single-agent baselines lag because the fixed tutor cannot recover chapter-specific keypoints without the corpus.

B. EduBench intersection and the capacity confound

On the EduBench intersection (n=1562), the capacity picture flips. Multi-agent retrieval improves EB12D mean +0.031 ($p<10^{-4}$), rubric coverage +0.186 ($p<10^{-4}$), factual-reasoning accuracy +0.055 ($p<10^{-4}$), and scenario adaptation +0.038 ($p<10^{-4}$), while cutting latency by 5,923 ms (−30%), tokens by 1,794 (−45%), and LLM calls by 0.49 per example (−25%). Classical RAG keeps a small edge on EduAlign (+0.040) and pedagogical-application (+0.008). The retrieval gate skips retrieval on every EduBench example (0% fire rate on 2,110 processed examples): the evidence score s_e is uniformly low for grading, emotional-support, and personalized-learning prompts where learner state matters more than corpus content. EduBench exercises the learner-, rubric-, and scenario-grounded branch of the policy; in this benchmark, the gate classifies examples as not requiring retrieval from an external corpus. Thus, on EduBench, the hybrid is essentially Multi-Agent (no retrieval) plus a cheap routing decision, which is why it nearly ties the retrieval-free variant (Δ_{EB12D} = +0.001, p=0.61) while beating classical RAG on the aggregate metrics and efficiency dimensions.

TABLE II

Head-to-head comparison on TutorEval (finished, n=834 identical examples) and on the EduBench intersection (n=1562 example_ids processed by all four mechanisms). Higher is better except latency, tokens, and LLM calls. Δ rows give (multi-agent retrieval − classical RAG) with paired bootstrap 95% CI and permutation p-value. The LLM-calls column records mean per-example backbone invocations; the gate in hybrid_fast reduces calls vs. classical RAG on EduBench. Best value in each column is bold.

Retrieval mechanism	Comp.	Tok-F1	EB12D	TE Hit	Rubric	EduAlign	Lat. (ms)	Tokens	LLM calls
TutorEval, n=834 identical examples (Qwen3.5-4B, thinking budget 512)									
Classical RAG	0.377	0.112	–	0.938	0.919	–	51 903	3 975	2.01
Multi-Agent (no retr.)	0.371	0.090	–	0.932	0.907	–	51 048	5 959	4.00
Single-Agent (no retr.)	0.360	0.086	–	0.900	0.867	–	55 257	6 466	2.00
Multi-Agent Retrieval (ours)	0.383	0.115	–	0.957	0.944	–	49 113	3 490	3.96
Δ (ours − Classical RAG)	+0.006	$+0.004^{*}$	–	$+0.019^{**}$	$+0.026^{**}$	–	−2,790	-484^{***}	+1.95
95% CI	–	[0.0003, 0.007]	–	[0.006, 0.032]	[0.007, 0.045]	–	[−7076, 1534]	[−558, −410]	–
EduBench intersection, n=1562 identical example_ids (Qwen3.5-4B)									
Classical RAG	0.367	–	0.367	–	0.705	0.166	19 828	4 020	1.97
Multi-Agent (no retr.)	0.398	–	0.398	–	0.889	0.130	13 916	2 216	1.47
Single-Agent (no retr.)	0.377	–	0.377	–	0.700	0.153	19 195	2 976	1.92
Multi-Agent Retrieval (ours)	0.398	–	0.398	–	0.891	0.126	13 904	2 226	1.48
Δ (ours − Classical RAG)	$+0.031^{***}$	–	$+0.031^{***}$	–	$+0.186^{***}$	-0.040^{**}	$-5{,}923^{***}$	$-1{,}794^{***}$	-0.49^{***}
95% CI	[0.027, 0.036]	–	[0.027, 0.036]	–	[0.174, 0.199]	[−0.069, −0.011]	[−6628, −5242]	[−1845, −1738]	[−0.52, −0.46]

$^{***}p<10^{-4}$, $^{**}p<0.01$, $^{*}p<0.05$; paired bootstrap 2000 resamples, permutation 10000 resamples.

TABLE III

Executed ablations on TutorEval (n=100, Qwen3.5-4B). Δ rows are (ablation − hybrid_fast baseline). (A) isolates the retrieval gate inside our retrieval-context policy; (B) confirms that the shared critic is an optional budget knob, not the source of our retrieval-side gains.

Condition	Tok-F1	TE Hit	Rubric	Correct	Tokens
baseline (hybrid_fast)	0.123	0.987	0.982	0.728	3 330
(A) gate off	0.123	0.992	0.986	0.731	3 449
Δ	+0.000	+0.005	+0.004	+0.004	+120
(B) critic off	0.135	0.981	0.983	0.727	2 405
Δ	+0.012	−0.006	+0.000	−0.000	−925

C. Ablations: attributing the hybrid's gains

Table III reports two executed ablations on a shared TutorEval slice (n=100). (A) Turning the retrieval gate off (force retrieval, i.e. making hybrid_fast retrieve on every example) nudges fixed-tutor scores by ≤ 0.005 on every metric but costs +3.6% tokens, so the gate is close to score-neutral and a clear efficiency win. (B) Disabling the critic – which is shared across all four mechanisms and therefore outside the scope of our innovation – keeps scores within ±0.006 of baseline on the scalar metrics but saves 28% of tokens, confirming the critic is an optional shared budget knob and not the source of our retrieval-side gains. A third ablation turning retrieval back on inside Multi-Agent (no retrieval) (non_rag_enable_retrieval, n=100) confirms that where retrieval is placed in the coordination policy—behind the router, not in front of it—matters as much as whether it runs at all; full numbers are released in docs/EXPERIMENTS.md.

Backbone replication. We re-ran all four mechanisms on Qwen3.6-27B-FP8 with the same endpoint, corpus, router, and response budget, pinning the model identifier through EndpointConfig.pinned_model. All eight 27B sessions completed. On TutorEval (n=200), both classical and hybrid essentially saturate the automatic tutor metrics (tutor-hit and rubric coverage ≥ 0.996), so the score deltas are indistinguishable from zero: Token-F1 −0.000 (p=0.94), tutor-hit +0.001 (p=0.59), rubric −0.002 (p=0.68). The efficiency result is not saturated: hybrid uses −992 tokens per example (95% CI [−1,136, −836], $p<10^{-4}$) and is −11,845 ms faster (95% CI [−13,553, −10,142], $p<10^{-4}$). On the 27B EduBench intersection (n=87), the 4B retrieval-context pattern repeats with larger margins: EB12D +0.032 ($p=2\times10^{-4}$), rubric +0.210 ($p<10^{-4}$), tokens −1,861 (−47%), and latency −22,070 ms (−55%). Thus the larger backbone does not erase the retrieval-policy result; it moves TutorEval from score-and-efficiency gains to efficiency gains at a metric ceiling, while EduBench keeps both.

V. Discussion

Coordination as a retrieval mechanism. The evidence supports a narrower claim than "agents are better tutors": multi-agent coordination can replace the retriever as the way tutoring context is built. On TutorEval, every 4B fixed-tutor score delta favors multi-agent retrieval at $p\leq0.05$; on the EduBench intersection, every reported delta except EduAlign and pedagogical-application favors it at $p<10^{-4}$. The planner, diagnoser, and rubric agents do the work classical RAG assigns to one embedded query: decide what to search for, which facts matter, and which learner state the tutor should condition on. The tutor and critic remain shared infrastructure.

The capacity "confound" flips on EduBench. Hybrid issues more LLM calls than classical RAG on TutorEval (3.96 vs. 2.01) because every example is chapter-grounded and the retrieval gate always fires. EduBench is different: the gate skips retrieval, calls fall from 1.97 to 1.48 per example ($p<10^{-4}$), tokens fall by 45%, and latency falls by 30%. Retrieval-context coordination is therefore not bought by uniformly spending more capacity. It is bought by replacing retrieve-then-read with a routed policy over the retrieval context: planning, diagnosis, rubric compilation, and retrieval compete for the same budget before the shared tutor runs.

VI. Limitations

Three constraints. (i) EduBench coverage. The intersections (n=1562 at 4B, n=87 at 27B) cover every example classical RAG finished at snapshot time, but not every example the coordination-heavy mechanisms reached. (ii) Human evaluation. We report automatic

metrics; a 4×25 stratified human-eval pack is packaged but not included. (iii) Scope. Two open-weight backbones, one shared corpus, English only, and benchmark outputs rather than learning gains. Both tracks regenerate end-to-end from the released configs.

VII. Conclusion

We compared classical RAG with three coordination-based retrieval alternatives under matched infrastructure on two open-weight backbones. On TutorEval (n=834), multi-agent retrieval improves Token-F1, tutor-hit, rubric coverage, and token use for the same tutor. On the EduBench intersection (n=1562), it improves EB12D, rubric coverage, latency, tokens, and LLM calls at $p<10^{-4}$. The 27B replication sharpens the story rather than changing it: TutorEval metrics saturate for both systems, but hybrid remains cheaper and faster; EduBench keeps both score and efficiency gains. In educational tutoring, multi-agent coordination can replace classical RAG as the retrieval mechanism–not by changing the tutor, but by changing the context the tutor receives.

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